Multiple Frame Manipulation in VR

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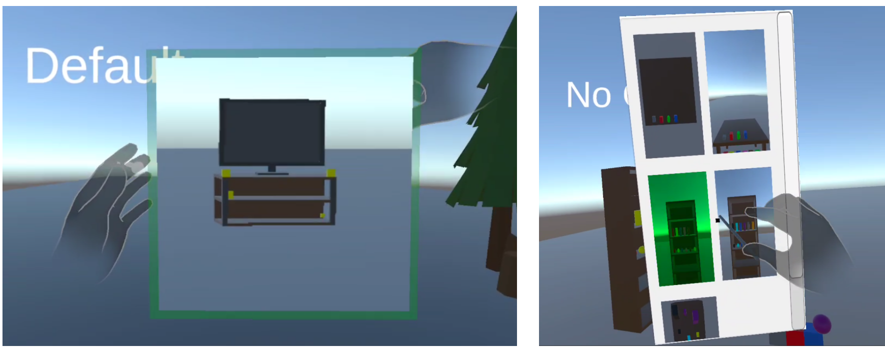

Fig. 1. Two-image teaser showcasing our system.

Many existing VR manipulation techniques require users to physically walk, reach, or adjust their viewpoint in order to reposition objects, which becomes inefficient for distant or occluded targets. This work explores a frame-based manipulation technique that enables users to create portable viewpoints and interact with objects through a 2D tablet interface using projection-based control. Our prototype supports viewpoint creation, frame switching, single-object manipulation, and lasso-based multi-object selection. We evaluated the system across three exploratory tasks and a comparative task against standard direct manipulation, using NASA–TLX workload measures, ease-of-learning ratings, SUS-based usability questions, and qualitative feedback. The results show that the frame-based approach reduces physical effort and enables interaction with distant objects but introduces substantial mental demand, sensitivity to small hand movements, and increased interaction complexity. Multi-object and long-distance operations benefited most from the technique, while precise within-frame manipulation remained challenging. These findings highlight both the potential and limitations of user-defined frame-based manipulation, motivating future refinement of depth control, feedback clarity, and interaction stability.

Additional Key Words and Phrases: 2D to 3D projection, frame-to-frame manipulation, in-frame manipulation, frame creation

1 INTRODUCTION

Many VR interaction techniques require users to physically move, reach toward objects, or adjust their viewpoint in order to reposition items in the scene. These constraints become inefficient when objects are far away, occluded, or positioned at uncomfortable angles, and are especially tedious when repetitive work is required. In this work, we explore a method that enables users to reposition objects from a mostly fixed location, independent of the objects' spatial placement. We focus on improving spatial coordination by providing clear visual feedback through frame-based cues that define the active manipulation region.

We address this problem by developing a prototype that allows users to create localized views of the scene and use these views as 2D interaction spaces for manipulating one or multiple virtual 3D objects. Within each view, users can reposition objects locally or teleport them between views, enabling scene-wide adjustments without physical navigation. The system supports both single-object manipulation and lasso-based multiple-object manipulation, allowing users to move individual items as well as self-defined clusters.

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Our approach builds on prior work in distant object manipulation. World-in-Miniature (WIM) techniques use a handheld miniature of the environment to act on distant objects via proxies, while tablet-based projection techniques map 2D input onto 3D transformations and enable view-dependent control from fixed viewpoints. Multiple-object manipulation techniques further show the value of treating sets of objects as a single unit for bulk operations. We adapt ideas from these lines of work to a setting where viewpoints are created by the user in situ and used as interactive frames embedded directly in the VR scene.

The goal of this exploration is to evaluate whether such frame-based manipulation can reduce physical effort and increase spatial accessibility while still supporting effective positional control. We implemented our technique on a Meta Quest 2 headset with hand tracking and conducted a user study that compares frame-based manipulation with direct manipulation, and examines how users experience viewpoint creation, within-frame manipulation, and frame-to-frame teleportation. We hypothesize that our approach can lower the ergonomic burden associated with repeated repositioning, while acknowledging that its view-dependent nature may introduce limitations in depth handling and fine-grained adjustment.

2 BACKGROUND

Regarding the related work, we will present some of the work that has been made in the XR community which relates to our domain dimensions, more specifically regarding stationary interaction with objects around the scene, the interaction mechanism with the objects, and the view-dependent 2D control using single or multiple perspective interactions. We do also consider the relevancy of multiple object selection, giving the user the possibility of moving full clusters from our frame to another.

One of the most influential ideas for interacting with distant objects in VR is the World-in-Miniature (WIM) technique introduced by Stoakley et al. . In WIM, the user holds a small, hand-sized version of the entire virtual environment, and any object inside this miniature can be moved as a proxy for its full-scale counterpart. This small model can be rotated freely, which helps reduce occlusion and gives the user multiple relevant viewpoints without needing to move around the actual scene. Because WIM works entirely through miniature proxies, the interaction stays indirect; users never manipulate the real-scale objects themselves. WIM is important for our work because it shows how additional viewpoints and alternative representations can make it easier to interact with objects at a distance, without requiring physical body move. At the same time, its reliance on scaled-down models and predefined perspectives suggests that other approaches, like working with real-scale, view-dependent representations, might support different kinds of tasks or offer different trade-offs. [3]

Bellgardt et al. investigate how 3D object manipulation in VR can be performed using traditional 2D input devices such as a mouse or graphics tablet, instead of 6-DOF controllers. Their technique maps 2D input to 3D transformations by projecting user actions onto the image plane of the VR camera, allowing users to translate and rotate objects through view-dependent control. Although their focus is on reducing fatigue and supporting setups where spatial controllers are impractical, their observations about 2D-to-3D projection—particularly depth ambiguity and the influence of camera angle—are highly relevant to our domain. Their work demonstrates how 2D projections can serve as an alternative interaction space for distant object manipulation without physical reach or constant viewpoint alignment. Conceptually, their system relies on fixed viewpoints and external 2D devices, whereas we explore similar projection principles using dynamically created viewpoints embedded directly in the VR environment. [1]

Beyond single-object techniques, VR research has also explored ways to manipulate groups of objects. Shi et al. present several group-based alignment techniques that allow users to select multiple objects and apply collective transformations, such as aligning items along shared axes or adjusting their spatial relationships. Their work highlights that VR scenes often require operations on clusters rather than isolated items, motivating interaction approaches that treat sets of objects as manipulable units. While their techniques rely on direct mid-air interaction and alignment widgets, they illustrate the broader relevance of multi-object manipulation in immersive environments. [4]

3 CONCEPT(S)

Given that the research question is "How can we enable users to specifically transform objects in a virtual scene without requiring physical movement, direct reach, or visual alignment with the objects' spatial positions?", the general conceptual idea is to manipulate 3D objects through a tablet artifact with a 2D interface, switching across multiple viewpoints initially created by the user. This relates to background work such as WIM and projection-based manipulation, where alternative viewpoints and 2D surfaces provide interaction access to distant objects.

Before interacting with the viewpoints, the user first creates them. Viewpoint creation is triggered through a bimanual gesture: the user performs a thumb-down, index-left, rest-fingers-curled pose with the right hand and pinches with the left. The size of the resulting frame determines the zoom level, where increasing or decreasing hand distance corresponds to zooming out or in. The potential primary benefit is simplicity, as the action resembles taking a picture with a camera. A liability is the dependency on precise hand recognition and pose detection, which can be affected by fatigue and occlusion. Since the gesture is not a one-to-one replication of real camera actions, memorability can also be a concern. This concept relates to WIM, which also highlights how access to alternative viewpoints can support distant object interaction. It leads to implementation through gesture detection and frame creation. Evaluation will examine whether users can perform these gestures reliably. A possible scenario is quickly capturing angles around large scenes without walking.

After defining the viewports, the user opens a tablet metaphor containing a menu of all frames. Opening happens with a palm-up, fingers-open pose; closing via a palm-down pose; and fixing the tablet position through a palm-up, fingers-curled pose. The main benefit is the familiarity of such poses, which are also used in other VR systems. The liability comes from pose detection varying across users. The tablet enables switching between frames, allowing fast access to different scene areas, though handling many frames at once can become challenging. This concept relates to WIM's multi-view idea, where different viewpoints provide access to different areas of the scene. It required implementation of pose-driven UI activation. Evaluation will look at whether switching frames reduces navigation effort or adds cognitive load. This concept is useful when users need quick access to multiple scene locations without teleportation.

When interacting with a frame, the user manipulates objects with a pen-based indirect manipulation metaphor. By hovering over an object's projection and pinching, the user can reposition it by dragging or by moving the pen to a target location and pinching. The benefit is repositioning objects without physical reach or line-of-sight alignment. The liability is reduced accuracy, since movement is based on 2D projection rather than depth-aware input. This concept relates directly to Bellgardt et al.'s 2D-to-3D projection technique, which also maps 2D input actions onto 3D object transformations. It required implementing ray–plane intersection logic. Evaluation will check how users manage depth and precision. This concept is useful for manipulating distant or occluded objects.

Regarding multiple-frame interaction, the user can switch between frames to manipulate the same object from different perspectives. This may help with depth-awareness issues, but requires additional frames, and switching can disrupt continuous interaction. This concept relates to both WIM, which reduces occlusion via alternative viewpoints, and to Bellgardt et al., whose depth ambiguity observations motivate the need for multiple perspectives. Its implementation required consistent object updates across frames, and evaluation will see whether users benefit from multi-view manipulation. It is useful when users need to verify an object's placement from several angles.

We also enable translating an object between viewpoints. This provides fast long-distance repositioning without physical effort or line-of-sight alignment. The liability is identifying the correct target frame when many frames exist, adding cognitive load, and losing continuous spatial feedback, which may affect the sense of control. This concept relates most closely to WIM, as both approaches support distant manipulation without requiring physical movement, although our method relies on user-defined frames rather than a miniature world. Implementation focused on object relocation logic, and evaluation will measure efficiency and user confidence. This is useful for reorganizing objects across large scenes.

The last concept, which is not the main focus but supports efficient repetitive work, is multiple-object selection and translation. The user creates a lasso with the pen inside a chosen viewport to select several objects. The benefit follows from the earlier local-manipulation advantages. The liability includes physical effort when drawing the lasso and difficulty avoiding accidental selections in crowded environments. This concept relates to Shi et al.'s work on group manipulation in VR, where sets of objects can be treated as a single manipulable unit. It required implementing lasso detection and multi-object updates. Evaluation will look at lasso accuracy and fatigue. This concept is useful when the task requires reorganizing several objects repeatedly.

4 IMPLEMENTATION

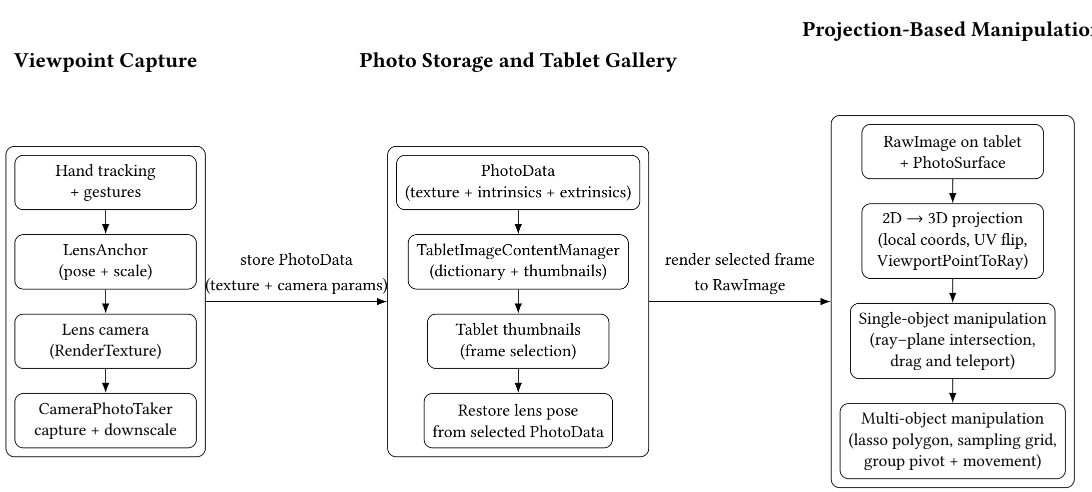

Fig. 2. Pipeline of the frame-based manipulation system, from viewpoint capture and storage to tablet-based projection and single/multi-object manipulation.

The implementation chapter explains the implementation of each of the features that the prototype contains, from short to long description, based on the feature relevance and originality. Using figure 2 as reference, we split our explanation into the 3 table sections above, having a conclusion at the end.

4.1 Viewpoint Capture

The viewpoint creation mechanism is implemented using a **LensAnchor GameObject** that represents the virtual camera used for capturing frames. The anchor's position, rotation, and scale are driven by hand tracking input, following the custom gesture described in the Concepts section. Gesture recognition is implemented using Meta's Hand Pose Detection system [2], where the right-hand pose defines the lens orientation and the left-hand pinch triggers the capture event. All gesture recognition based interactions are implemented with identical mechanism, therefore not worth mentioned.

When the user performs the capture gesture, the **CameraPhotoTaker** component records the current render texture from the lens camera, formats a downscaled copy for use as a thumbnail on the tablet, and extracts the camera's intrinsic and extrinsic parameters. These values are stored in a **PhotoData** structure, which represents a single viewpoint and is later consumed by the tablet interface. The system also provides lightweight visual feedback (a brief green highlight on the lens), but this is secondary to the capture pipeline.

A practical challenge was ensuring consistent pose detection under different hand orientations and distances, especially when the lens was scaled or rotated quickly by the user.

4.2 Photo Storage and Tablet Gallery

Given saved render textures and camera parameters in each **PhotoData**, these packets are forwarded to the **TabletImageContentManager**, which stores them in a dictionary and populates the tablet gallery with thumbnail instances. Each thumbnail displays the downscaled texture associated with its **PhotoData**, giving the user a quick overview of all captured viewpoints.

When a thumbnail is selected, the manager retrieves its corresponding **PhotoData** and reconstructs the view by repositioning the shared **lensAnchor** and applying the saved intrinsic camera parameters to the **lensCamera**. Instead of creating a new camera for every frame, the system reuses this single hidden camera, temporarily moving it into the stored pose and rendering its output onto a `RenderTexture` shown on the tablet. This makes it possible to display a live render of the selected viewpoint rather than a static screenshot.Important to mention is that the same physical camera component is used both for capturing the initial viewpoint and for reproducing it later on the tablet.

A notable implementation detail is that all frames can be refreshed after object manipulation. The manager iterates through every stored **PhotoData**, repositions the shared camera to the saved pose, and re-renders into the photo's texture. This ensures that every thumbnail and full-view image remains consistent with the current state of the 3D scene.

4.3 Projection-Based Manipulation

The most complex section of our implementation is the Projection-Based Manipulation, given that we have pressed on a thumbnail and the tablet now showcases only the `RawImage` representing the particular thumbnail. The `RawImage` contains the **PhotoSurface** class, responsible for 2D interaction with 3D real objects within the frame.

The core of this interaction is the mapping from 2D pointer positions on the `RawImage` to 3D rays in the virtual scene. When the user moves the pen over the tablet, we first convert the screen-space pointer position into local coordinates of the RawImage and then normalize these to UV space. Because the captured textures are flipped to match the tablet layout, the UVs are inverted horizontally and vertically before being passed to the **lensCamera** via `ViewportPointToRay`. This gives us a world-space ray that is consistent with the original viewpoint and can be used to raycast against objects in the scene. The inverse mapping (from world positions back to RawImage coordinates) is also implemented and used to place certain UI elements, such as the multi-object pivot indicator, directly in image space.

Single-object manipulation combines this projection with a dynamically defined interaction plane. When the user clicks on an object in the frame, we select it as the current target and provide visual feedback through a glow effect. During dragging, we construct a plane that passes through the object's current world position and is orthogonal to the viewing direction of the lens camera. For each pen movement, the projected ray is intersected with this plane, and the intersection point becomes the new position of the object. This effectively constrains movement to a plane that is view-aligned, which gives a 2D tablet drag the effect of sliding the object across a planar surface in 3D. The same mechanism is used for teleportation when the user taps on empty space with an object selected: instead of continuous updates, the object is moved in a single step to the plane–ray intersection.

Multi-object manipulation builds on the same projection principle but uses a lasso-based selection in image space. When the user drags without a single object selected, the system records the 2D trail on the RawImage as a polygon and visualizes it as a polyline. Once the gesture ends, we close the polygon, check that it is not self-intersecting, and sample a grid of points inside its bounds. Each grid point is projected through the same function into a 3D ray, and we accumulate which objects are hit inside the lasso region as well as across the entire RawImage. Objects whose inside-hit ratio exceeds a threshold are added to the selection set. These objects are then reparented under a common pivot, placed at the centroid of the group, and can be translated together using the same projection-and-plane logic as for single-object dragging.

4.4 Conclusion

To conclude the implementation chapter, we briefly outline the software and hardware used in the prototype. The system was developed in Unity 2022 using C# as the primary programming language. For hand tracking and gesture detection we relied on the Meta Interaction SDK. The only external UI-related library used was the Unity UI Extensions package [5], specifically for rendering polylines during the lasso selection process; this library was used purely for visualization, not for any of the mathematical logic behind selection or clustering. The prototype was deployed and tested on a Meta Quest 2 headset.

5 EVALUATION

5.1 Task Design

The evaluation is divided into two main tasks, where Task 1 consists of three exploratory subtasks and Task 2 is a comparative task. Task 1 focuses on understanding how users interact with individual features of the system, while Task 2 compares our complete prototype against a well-established baseline technique.

Task 1.1 evaluates how users create viewpoints and how they navigate and inspect the resulting thumbnails on the tablet menu. Task 1.2 requires participants to replicate a predefined structure using objects distributed across two tables and two shelves, testing only frame switching and single-object manipulation within the tablet interface. Task 1.3 asks users to move a specific group of objects (one object type from a mixed cluster) from one table to another, evaluating frame switching and the lasso-based multiple-object selection mechanism.

Task 2 involves sorting colored cubes into a trash bin with four color-coded openings. Participants first complete the task using standard direct manipulation, physically reaching around the bin and occasionally kneeling to place cubes into the correct opening. They then repeat the same task using our full prototype, excluding the multiple-object selection feature so that both conditions consist of identical repeated single-object actions.

This design allows us to compare two types of constraints: the *physical and ergonomic demands* of direct manipulation (walking, reaching, bending) and the *mental and interaction overhead* introduced by the frame-based system (creating viewpoints, switching frames, and operating the tablet interface). Focusing solely on single-object actions ensures that the comparison remains fair and isolates the core manipulation mechanism without introducing advantages from bulk operations.

5.2 Conditions/Independent Variables

Only Task 2 introduces a comparative condition. The independent variable in this task is interaction technique, with two levels:

- Direct Manipulation
- Frame-Based Manipulation (our prototype)

5.3 Apparatus and Implementation

All tasks were performed on a Meta Quest 2 headset running our Unity 2022 prototype in standalone mode. The system used hand tracking exclusively for all interactions, including viewpoint creation, tablet activation, single-object manipulation, and lasso-based multi-object selection. No external controllers or additional hardware were required.

The virtual environments for the tasks were built from simple geometric structures and varied only in layout and object placement:

- **Task 1.1:** A scene containing four spatially separated clusters at different distances and scales, used to evaluate viewpoint creation under diverse spatial conditions.
- **Task 1.2:** Two tables and two shelves populated with basic objects. Participants reconstructed a target arrangement on the main table using five predefined viewpoints, focusing on frame switching and single-object manipulation.
- **Task 1.3:** Two tables, where one held mixed-color cubes. Participants selected all cubes of a target color using the lasso tool and moved them to the other table.
- **Task 2:** A table with colored cubes and a trash bin containing four color-coded openings. Participants completed the sorting task first via direct manipulation and then using the frame-based system (with multi-object selection disabled).

All tasks were performed while standing within a safe play area of approximately $2\,\text{m} \times 2\,\text{m}$. Interaction features were selectively enabled or disabled depending on the requirements of each task.

5.4 Participants

A total of **8** participants participated in the study, 4 with moderate experience and 4 experienced. Participants were recruited through convenience sampling at the university and volunteered without compensation.

Before the study, all participants were informed about the purpose of the evaluation, the interaction techniques involved, and their right to withdraw at any time. No personal data beyond self-reported VR experience was collected.

5.5 Procedure

The study followed a linear sequence of four tasks. Participants did not receive a formal training phase; instead, each task was briefly explained immediately before it was performed. The explanation described only the goal of the task and the interaction elements required to complete it. Participants then executed the task on their own, and clarifications were given only if they became stuck.

The session began with Task 1, which consisted of three exploratory subtasks. In Task 1.1, participants created viewpoints around several scene clusters and inspected the resulting thumbnails on the tablet. In Task 1.2, they replicated a target object arrangement using the predefined viewpoints and single-object manipulation. In Task 1.3, they used the lasso-based selection mechanism to select all objects of a specific type and move them to the other table.

Task 2 was a comparative condition and was always performed in a fixed order. Participants first completed the sorting task using direct manipulation by physically placing cubes into the matching openings of the trash bin. They then repeated the same task using the frame-based system (with multiple-object selection disabled), ensuring that both conditions involved identical single-object actions.

After each task or subtask, participants completed the corresponding questionnaire. The full study session, including all tasks and questionnaires, lasted approximately 30 to 40 minutes per participant. Participants were encouraged to proceed at their own pace, ask questions when needed, and take breaks if required.

5.6 Evaluation Metrics

To assess the usability and workload of the interaction techniques, we collected a mix of subjective quantitative measures, qualitative feedback, and simple behavioural indicators.

5.6.1 Task 1.1–1.3 (Exploratory Feature Evaluation). These subtasks focus on understanding individual components of the system. For each subtask, we collected:

- **Subjective workload ratings** using 7-point NASA–TLX style items: mental demand, physical demand, temporal demand, perceived performance, effort, and frustration.
- **Ease-of-learning ratings** on a 1–7 scale for subtask-specific interactions (e.g., frame creation, switching between frames, single-object manipulation, multi-object lasso selection).
- **Behavioural indicators**, such as the number of photos taken (Task 1.1) or the number of lasso attempts (Task 1.3), providing insight into how often actions required retries.
- **Qualitative responses**, where participants described what they liked or disliked about each interaction method, as well as any challenges they encountered.

These metrics were chosen to capture how understandable, demanding, and controllable each feature felt to users, since the subtasks do not involve a comparison against a baseline.

5.6.2 Task 2 (Comparative Evaluation). Task 2 evaluates the full prototype in comparison with standard direct manipulation. For this task, we collected:

- **Comparative subjective ratings** on a 1–7 scale assessing ease of sorting cubes, physical comfort, and the effectiveness of each technique when objects were occluded or positioned awkwardly.
- **Usability ratings** using a set of 1–5 Likert items inspired by the System Usability Scale (SUS), evaluating perceived complexity, consistency, learnability, confidence, and overall integration of system functions.
- **Qualitative comparative feedback**, where participants explained perceived differences between direct manipulation and frame-based interaction, and identified strengths or limitations of each approach.

Together, these metrics provide insight into user effort, perceived control, usability, and the relative advantages of the frame-based technique compared to traditional interaction.

6 RESULTS

6.1 Task 1.1: PhotoLens and Canvas

Measure	Mean
Mental Demand	3.75
Physical Demand	3.13
Temporal Demand	2.13
Performance (1=Good,7=Poor)	4.13
Effort	5.00
Frustration	2.75
Learning: PhotoLens	3.88
Learning: Canvas	5.38
Learning: Combined	4.75

Table 1. Task 1.1: NASA-TLX and ease-of-learning scores.

All participants completed Task 1.1 and successfully captured four photos with the PhotoLens. The PhotoLens was perceived as intuitive but limited by imprecise hand–lens alignment, an overly strong zoom, and high sensitivity to small hand movements. The Canvas was also seen as intuitive and familiar, although its limited size, the number of required gestures, and the need to close it before activating the PhotoLens were experienced as drawbacks. Across both interactions, the sensitivity to hand position and distance reduced perceived stability and accuracy.

6.2 Task 1.2: Frame Switching and Single-Object Manipulation

Measure	Mean
Mental / Physical / Temporal	5.75 / 3.38 / 2.75
Performance / Effort / Frustration	4.00 / 5.38 / 4.50
Learning: Switching Frames	4.00
Learning: Move Within Frame	3.13
Learning: Move Between Frames	4.38
Learning: Combined	3.50
Position Understanding	3.38
Worthwhileness of Extra Steps	4.13

Table 2. Task 1.2: Workload, ease-of-learning, and evaluation of the frame-based interaction.

Participants appreciated the multiple-viewpoint approach for enabling inspection of objects from different angles and found the concept intuitive once learned. However, they reported that the frame-based workflow required many steps, navigation between frames was not always intuitive, and both pen-based manipulation and object placement were highly sensitive to small movements, making precise adjustments difficult.

6.3 Task 1.3: Multi-Object Lasso Selection

Measure	Mean
Mental / Physical / Temporal	2.50 / 2.13 / 2.75
Performance / Effort / Frustration	4.50 / 2.88 / 2.38
Learning: Polyline Selection	6.00
Learning: Multi-Frame Move	4.13
Lasso Feedback Clarity	5.25
Selection Understanding (1=Easy,7=Difficult)	2.88

Table 3. Task 1.3: Workload, ease-of-learning, and lasso feedback evaluation.

Participants generally found the lasso-based condition intuitive, fast, and enjoyable, and they appreciated being able to select multiple objects at once with relatively little effort. Criticisms focused on delayed or unclear feedback, difficulty when moving selected objects, and the presence of a distracting guidance cube, which sometimes made the interaction feel slow or hard to control.

6.4 Task 2: Overall Usability and Frame-Based Sorting

Measure	Score
SUS Scores (Participants 1–8)	67.5, 57.5, 65.0, 55.0, 57.5, 57.5, 50.0, 60.0
SUS Mean	**58.75**
Ease of Assigning Cubes (1–7)	4.13
Physical Comfort During Sorting (1–7)	4.13

Table 4. Task 2: System Usability Scale (SUS) and task-specific ratings.

For Task 2 specifically, participants rated both the ease of assigning cubes and the physical comfort of the sorting task at a mean of 4.13 on a 7-point scale. Qualitative feedback highlighted that the frame-based method reduced physical movement and could be faster once frames were prepared, but it also introduced substantial interaction complexity and mental demand compared to direct manipulation, resulting in a trade-off between physical convenience and cognitive effort.

7 DISCUSSION

Our aim was to reduce physical effort by enabling users to manipulate objects from a stationary position through frame-based interaction. The results show that this benefit was achieved to some extent—participants appreciated not having to walk or reach to handle distant or occluded objects. However, this ergonomic gain was consistently overshadowed by the increased mental effort, sensitivity issues, and interaction complexity introduced by the system. The mid-range workload ratings and a SUS score of 58.75 indicate that the overall trade-off between reduced physical strain and added cognitive burden was not favourable.

The strongest limitations appeared in within-frame manipulation using the pen-based projection. Users struggled with unstable control, sensitivity to small hand movements, and difficulty understanding how depth was handled. These issues closely reflect observations from projection-based manipulation research, where view-dependent control introduces depth ambiguity unless additional mechanisms are provided. In contrast, frame-to-frame transfers and multi-object lasso selection were better received. These interactions aligned more naturally with the frame metaphor and required less precise moment-to-moment control, similar to how WIM and related multi-view techniques support coarse but efficient object repositioning. Participants particularly valued the ability to move clusters of objects and to reposition distant elements without navigating the scene physically.

Overall, the findings suggest that our contribution—portable, user-defined viewpoints used as portals for remote manipulation—has conceptual merit. However, the current interaction design imposes unnecessary complexity, and the system performs best when used for operations that benefit from remote access rather than precise within-frame manipulation. Reducing the number of interaction steps, simplifying mode transitions, improving feedback, and stabilising object movement are necessary steps for making the technique truly competitive with direct manipulation.

8 CONCLUSION AND FUTURE WORK

This work introduced a frame-based technique for stationary object manipulation in VR, inspired by concepts from World-in-Miniature and projection-based control. The system successfully reduced physical effort and enabled interaction with distant or occluded objects, validating the idea that portable viewpoints can support remote manipulation. However, the high mental effort, sensitive input behaviour, and difficulty with depth control limited overall usability. Participants benefitted most from long-distance transfers and multi-object operations, suggesting that the approach is better suited for coarse spatial adjustments than for precise, fine-grained tasks.

Future work should prioritise simplifying the interaction pipeline and stabilising within-frame manipulation through filtering, snapping, or constrained movement. A major remaining challenge is depth control: because interaction occurs on a single view-aligned plane, users must rely on switching frames to correct depth placement. Potential improvements include dual-plane manipulation, automatic depth "rails," or intuitive pinch-based gestures that allow explicit depth adjustment without abandoning the frame metaphor. Integrating these refinements may allow the technique to retain its strengths—remote reach, multi-object efficiency, and reduced physical navigation—while mitigating the cognitive overhead that currently limits its practicality. More broadly, future systems may benefit from hybrid approaches that combine the remote access of frame-based interaction with lightweight local viewpoint adjustments or adaptive feedback to support tasks requiring higher precision.

Youtube link: https://www.youtube.com/watch?v=F7R_Q_KAO7U

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